

SuperMuSR Build Update

2026 06 02

Recent timeline estimates

Hi Mark, hope you don't mind me CC-ing people who have just asked me a similar question!

The timeline from my perspective is below, dictated by milestones such as beam time.

Francesco may have a have a different, pragmatic timetable that weighs in other priorities. He is currently compiling a timeline so please take this as tentative:

I must have all modules assembled and tested by beam cycle ending 19 Dec 2026. That is the next 6 months. The modules are the items seen as A A B C below. There are 120 modules to make, plus spares. Currently I do not have funding for 24 of these modules - those for the 8 short staves) . This week detector systems group (CC Jeff) allocated the funds for spares as a long as it is ASAP. When might I get a decision on short-stave funding?

For the R106 pre-build we aim to have the skeleton barrel pair (similar to the top left image) to you for trial fitting around July. That will be the metal skeleton with some 3d printed parts that represent the modules. @Prince, James (STFC,RAL,ISIS) might need to confirm the delivery time for those 3D parts. This is a mechanical test in R106. The “will it fit” test. You can have the barrel earlier but I don’t think R106 will be ready for it. So, before July it is going to Andy Church for thermal testing and Zoe Tam for load testing.

The modules (see A A B C below) are being built from mid-june onwards. I say mid June as that’s when the electronics (PCBs) arrive. Everything else is here, and Neil has made a decent stockpile of tiles ready! Scintillator tile production will continue in parallel, and will be completed June->Dec. Hopefully earlier.

All modules should be completed by Dec 18th, the earlier the better. I need lots of modules in-hand in September to start beam testing. We are reserving RIKEN Port 3 for this, where I am building a "quick swap" test station.

@Zuddas, Francesco (STFC,RAL,ISIS) asked how many lots its 😊 . Between 32 and 64 modules is what I have in mind. The modules go together quickly. I think this is achievable.

Putting the modules on the stave, and the stave in the barrel can commence as soon as:

1. The first modules are done and tested
2. I have the barrel back from R106.

This can happen in parallel too, as we go. So I hope everything is done by Christmas. Or at least the modules are done by Christmas and we load it all into the barrel early 2027.

I hope that makes sense.

Dan

Short answer:

1. Scintillator tiles = until Dec
2. Module build = June to Dec.
3. Barrel to R106 = July to September
4. Detector Build = Dec to Feb.

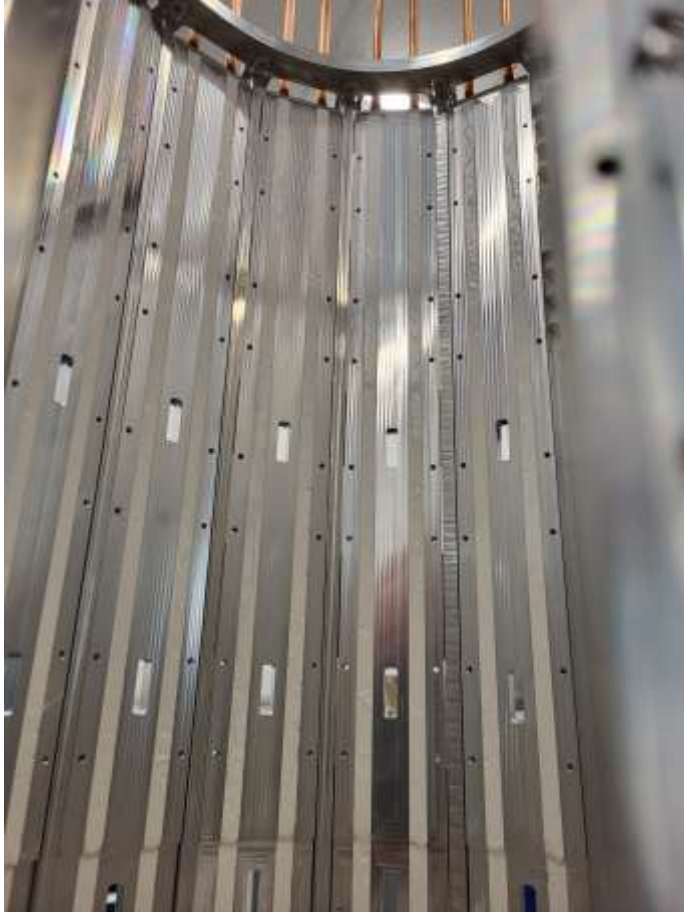
Update

- Detector electronics due around now. They are with NI who are doing their (extensive) QA checks/ reporting.
- Next steps are clear. Some discussion about module build while tiles also building tiles. Suspect this can be done concurrently.
- Sent out timeline from my point of view (my milestones). Beam time really important – planning Riken ‘quick test’ setup.
- **Risk** – need approval for remaining spend for short staves. Takes about 4 months from PO to in-hand. (company are extremely quick), but we are out of time to wait. £130000. DSG has put funds aside for spares, but need to buy full quantity for price-break.

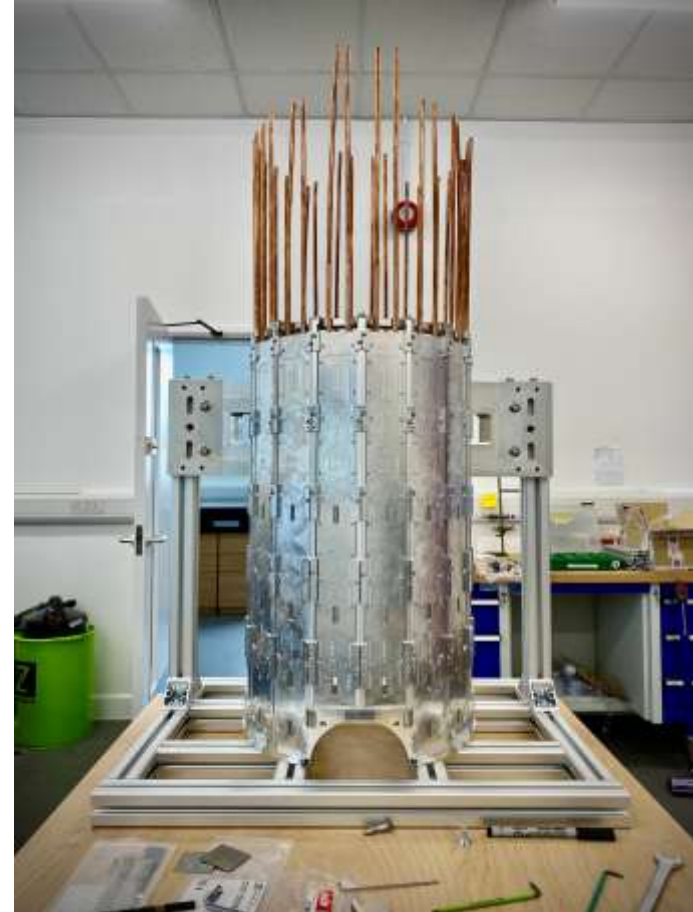
SuperMuSR Build (RAL Tech – Neil / Will)



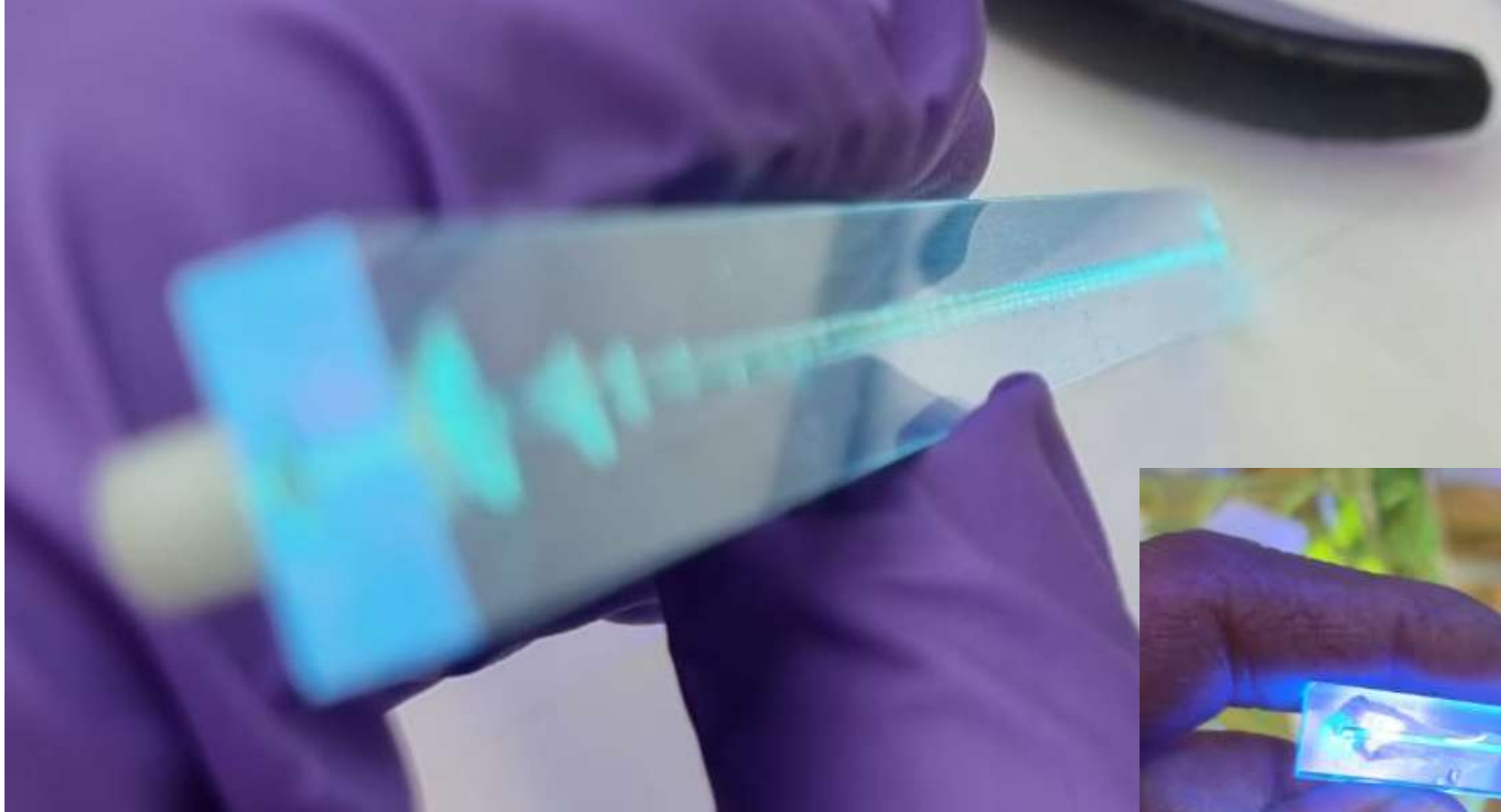
SuperMuSR Build (RAL Tech – Neil / Will)



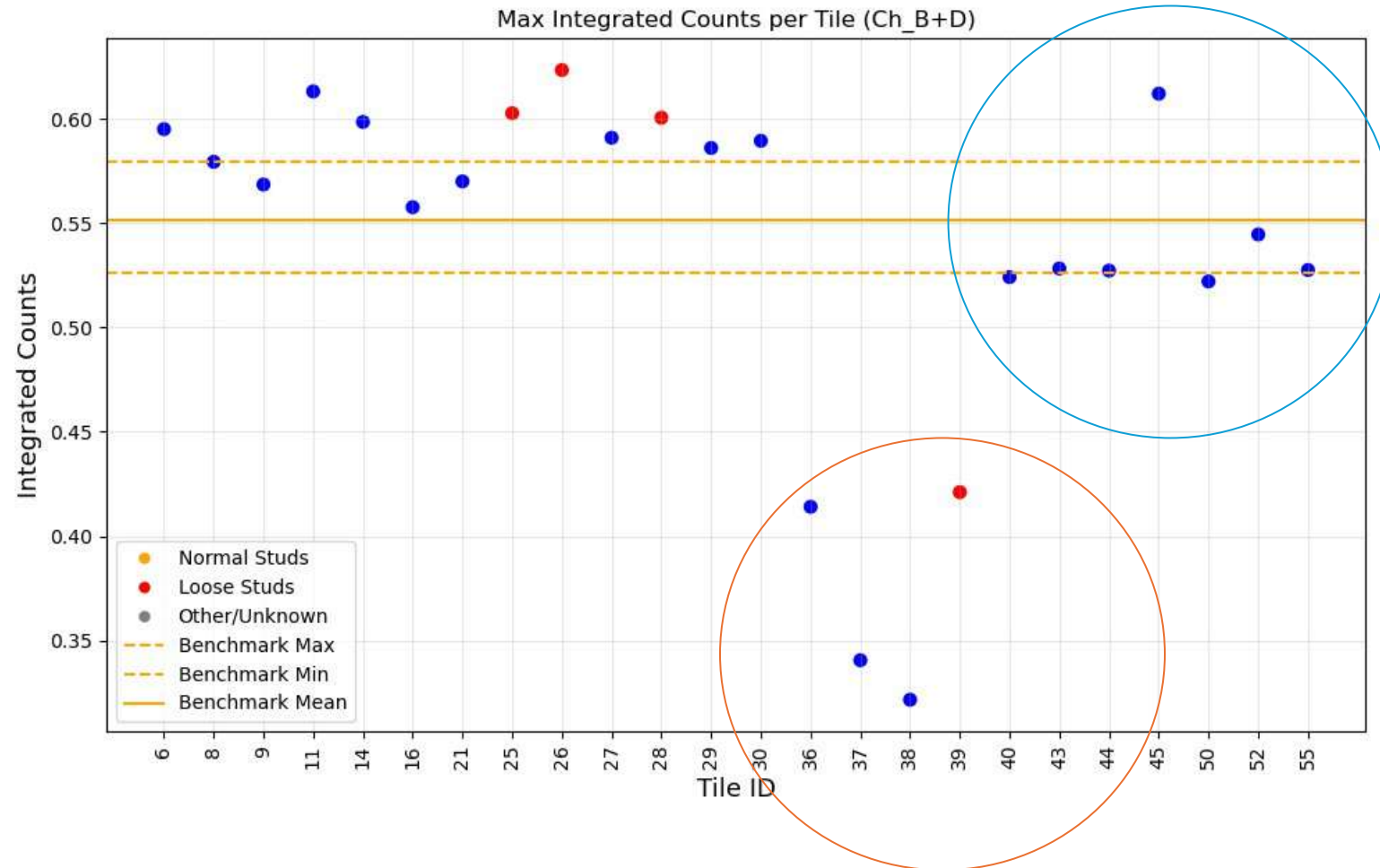
SuperMuSR Build (RAL Tech – Neil / Will)



Drop in tile performance. Suspect WSF encapsulation has become poor for optical coupling (RAL Tech – Neil / Will)



Drop in tile performance under investigation



Problems come and go



One stave of 16 (per barrel) , loaded into skeleton framework

